

FOSSIL INVERTEBRATES

Invertebrates, animals without a solid internal skeleton, are among the most common fossils found. They first appeared in the Precambrian, but it is only in Early Cambrian times that complex invertebrates such as trilobites become numerous in the fossil record. Fossils of invertebrates such as arthropods, molluscs, brachiopods, echinoderms, and corals are especially common, because they had hard external structures and lived in the sea, where most fossil-bearing rocks form.



tube-like compartment

sheet-like structure

CHEILOSTOME BRYOZOAN *Biflustra* sp.

Found in Cenozoic rocks, the *Biflustra* genus is extant. These bryozoans have minute compartments, which house zooids – the soft-bodied individuals of a colony.



SERPULID WORMS *Rotularia bognoriensis*

Found in Jurassic to Eocene rocks, *Rotularia* is a genus of serpulid worm. Like all serpulids, each worm protected its soft body by producing coiled tubes made of calcium carbonate.



“TUNING FORK” GRAPTOLITE *Didymograptus murchisoni*

A graptolite (extinct colonial invertebrate) with two stipes (branches), *Didymograptus* is found in Ordovician rocks. It grew from 2–60 cm (¾–23½ in) in length.



STROMATOPOROID

Stromatopora concentrica

Found in rocks ranging from the Ordovician to the Permian, often in reef limestone, the fossil sponges in this group were made of porous, calcium-rich tubes.



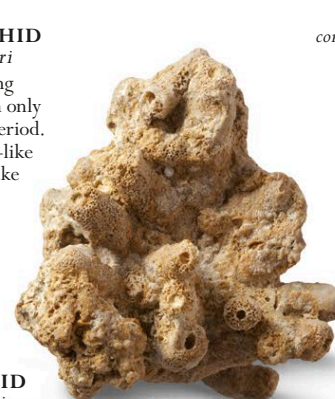
TREPOSTOME BRYOZOAN *Diplotrypa* sp.

A bryozoan from Ordovician strata, *Diplotrypa* was a small invertebrate, not unlike coral, that lived in dome-shaped colonies.



LACE CORAL

Schizoretepora notopachys
A lace coral, *Schizoretepora* is found in Eocene to Pleistocene strata. It lived on rocky seabeds.



ARCHAEOCYATHID *Metaldetes taylori*

These reef-building organisms are known only from the Cambrian Period. *Metaldetes* had a cup-like structure, not unlike that of a coral.

CALCAREOUS SPONGE *Peronidella pistilliformis*

Characterized by needle-like spicules (in the form of calcite) fused together, *Peronidella* is found in Triassic and Cretaceous rocks.



BRANCHING BRYOZOAN *Constellaria* sp.

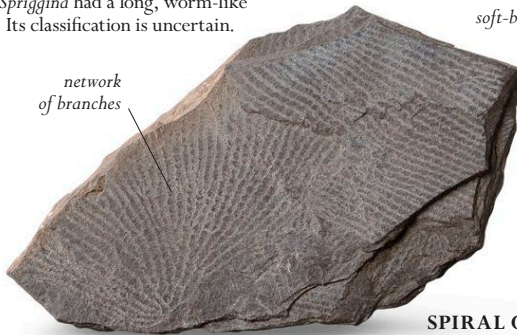
A bryozoan that built branching colonies on the seabed, *Constellaria* occurs in Ordovician strata.



SPRIGGINA

Spriggina floundersi

A very early fossil found in Ediacaran rocks, *Spriggina* had a long, worm-like body. Its classification is uncertain.



network of branches

BRANCHING GRAPTOLITE *Rhabdinopora socialis*

Until recently, this graptolite was called *Dicyonema*. It had numerous thin, radiating stipes, and is from Ordovician strata.

thecae housed soft-bodied individuals of the colony

single, curved branch

SPIRAL GRAPTOLITE *Monograptus convolutus*

A single branch, with thecae (cup-like structures) on one side, characterizes *Monograptus*, which is found in Early Silurian strata. *M. convolutus* had unusual coiling.



corallite